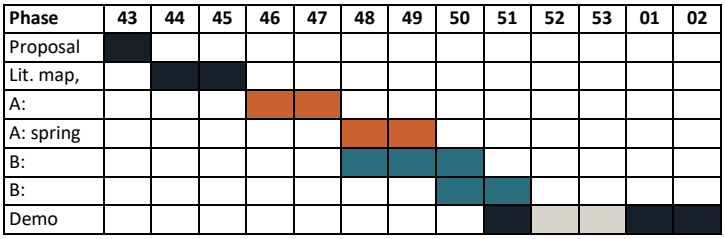
MDM2 – Case Study: Assistive Robotics for Improvement of Life Quality
One-Page Proposal

Team	Team 7
Members	Vedant Joshi, Navya Nagendra, Harshil Safi
Project Title	Simulation-based design of wrist exoskeleton assistance for post-stroke wrist-extensor weakness: optimal control vs. reinforcement learning
GitHub Repository URL	https://github.com/Rationaliser/wrist-exo-assist-sim
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Industrial Application (target domain/use-case)	Wrist exoskeletons and dynamic orthoses for stroke rehabilitation and daily assistance: sizing actuators (spring vs. motor) and developing controllers in simulation before building hardware.
Keywords (3–6)	Wrist exoskeleton · stroke · musculoskeletal simulation · optimal control · reinforcement learning · MyoSuite
Submission Date (YYYY-MM-DD)	2026-10-21
Gant Chart	Calendar weeks (KW), Oct 2026 – Jan 2027, aligned with course deadlines  Dark: shared · Orange: Route A (Moco) · Teal: Route B (RL) · Grey: holiday buffer

1) Problem Statement & Measurable Outcomes (3–4 sentences)	After a stroke, weak wrist extensors make the wrist drop, so everyday tasks such as lifting and holding a cup fail. Developers lack a cheap way to know how much wrist assistance is needed, and whether a spring is enough or a motor is required, before building hardware. This project answers that in simulation and compares an optimal plan with a learned controller. Outcomes: **K1** Extensor strength threshold (% of max. muscle force) where lifting and holding the cup fails **K2** Minimum wrist assist torque (Nm) that restores the task, and whether a spring is enough **K3** Wrist-muscle effort reduction (%) compared with no assistance **K4** Success rate (%) of the RL policy vs. the Moco plan with a heavier cup, pushes and fatigue
2) Motivation & Assistive Relevance (2–3 sentences)	Stroke survivors rank recovery of arm and hand function among their top rehabilitation priorities; a stable wrist brings back drinking, eating and carrying. Developers get actuator requirements before prototyping, and open-source tools make simulation-first design possible.
3) Related Work Snapshot (2–3 key references)	**KhalilianMotamed Bonab et al. 2024** (Robotics): OpenSim simulation of an elbow–wrist exosuit; healthy subjects, no weakness, no learned control. **Caggiano et al. 2022** (MyoSuite): wrist/hand models with exoskeleton, sarcopenia, fatigue options. **Chavarrías et al. 2025** (arXiv): RL exoskeleton controller trained in MyoSuite; knee, not wrist. **Gap:** we found no study that sizes wrist assistance for a weakened, post-stroke wrist or compares an optimal plan with a learned controller.
4) Method & Feasibility (≤6 sentences)	**Route A (primary):** OpenSim Moco with the MoBL-ARMS upper-limb model predicts lifting and holding a cup; wrist-extensor strength is reduced step by step until the task fails. **Assistance:** a spring and an ideal motor at the wrist, each optimised for minimum muscle effort. **Route B (secondary):** a PPO policy on MyoSuite's myoHand learns the cup task, guided by the Moco motion; full hand control is a stretch goal. **Data:** no human data; open models, checked against published wrist muscle patterns. **Feasibility:** open-source tools, validated upper-limb models, and a fallback if RL training fails. **Artifacts:** GitHub repository, plots, report.
5) Milestones & Timeline (short table/list)	**P1** Proposal: 21 Oct · **P2** Lit. map, method outline, baseline sim, bibliography report: 4 Nov **P3** Methodology: 18 Nov and 2 Dec · **P4** Demo, experiments and analysis: 16 Dec **P5** Final presentation and report: 13 Jan 2027
6) Risks & Ethics (1–2 sentences)	Simulation only: no human subjects or personal data, all sources cited, and results reported as relative findings, not clinical claims. Main risks are solver or RL training failure and model mismatch, handled by a reduced wrist model, a reference-guided reward and an agreed fallback.